



BMDS2133 IMAGE PROCESSING

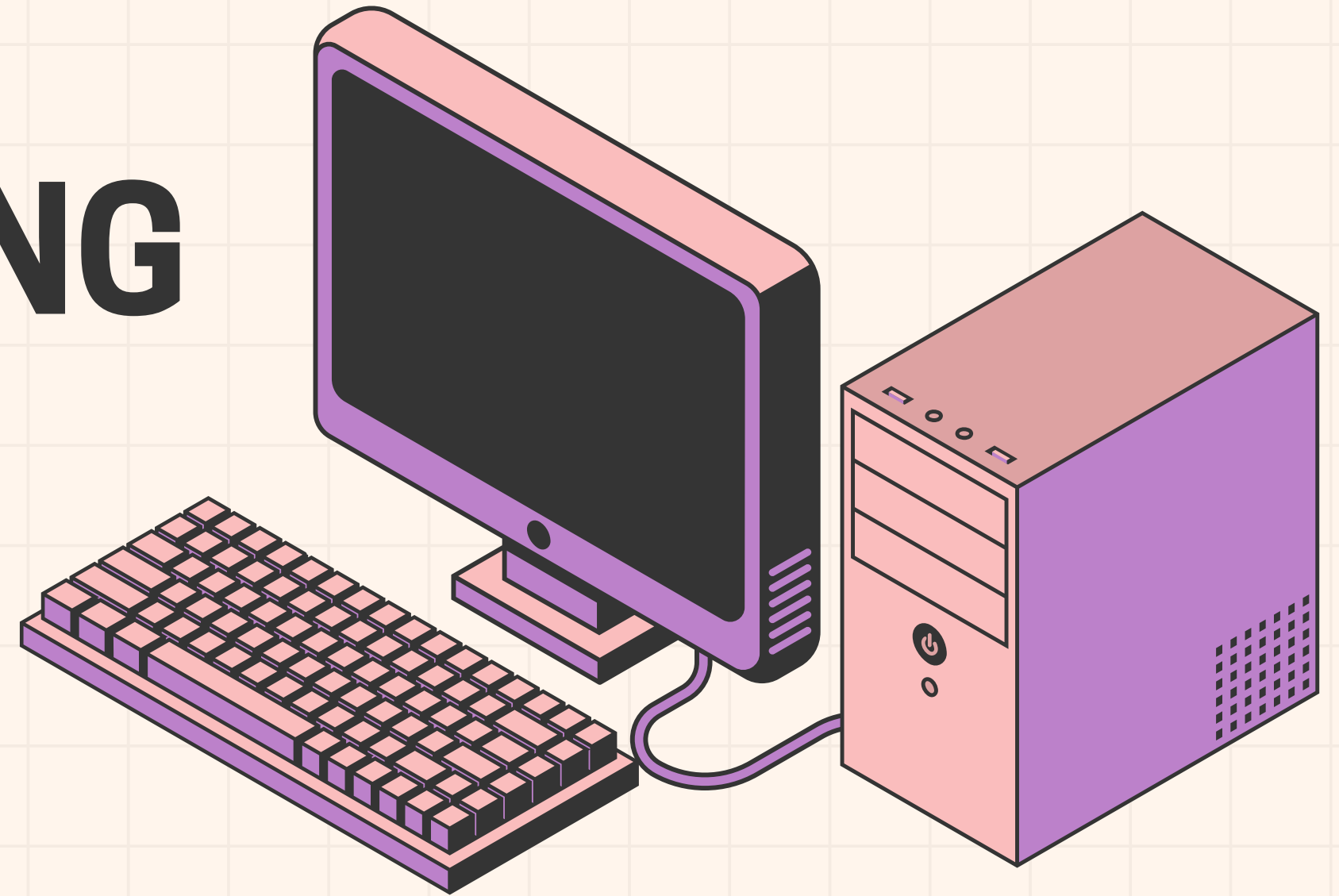
Group Member:

Lim Feng Zhi

Low Zi Qing

Ng Chiao Han

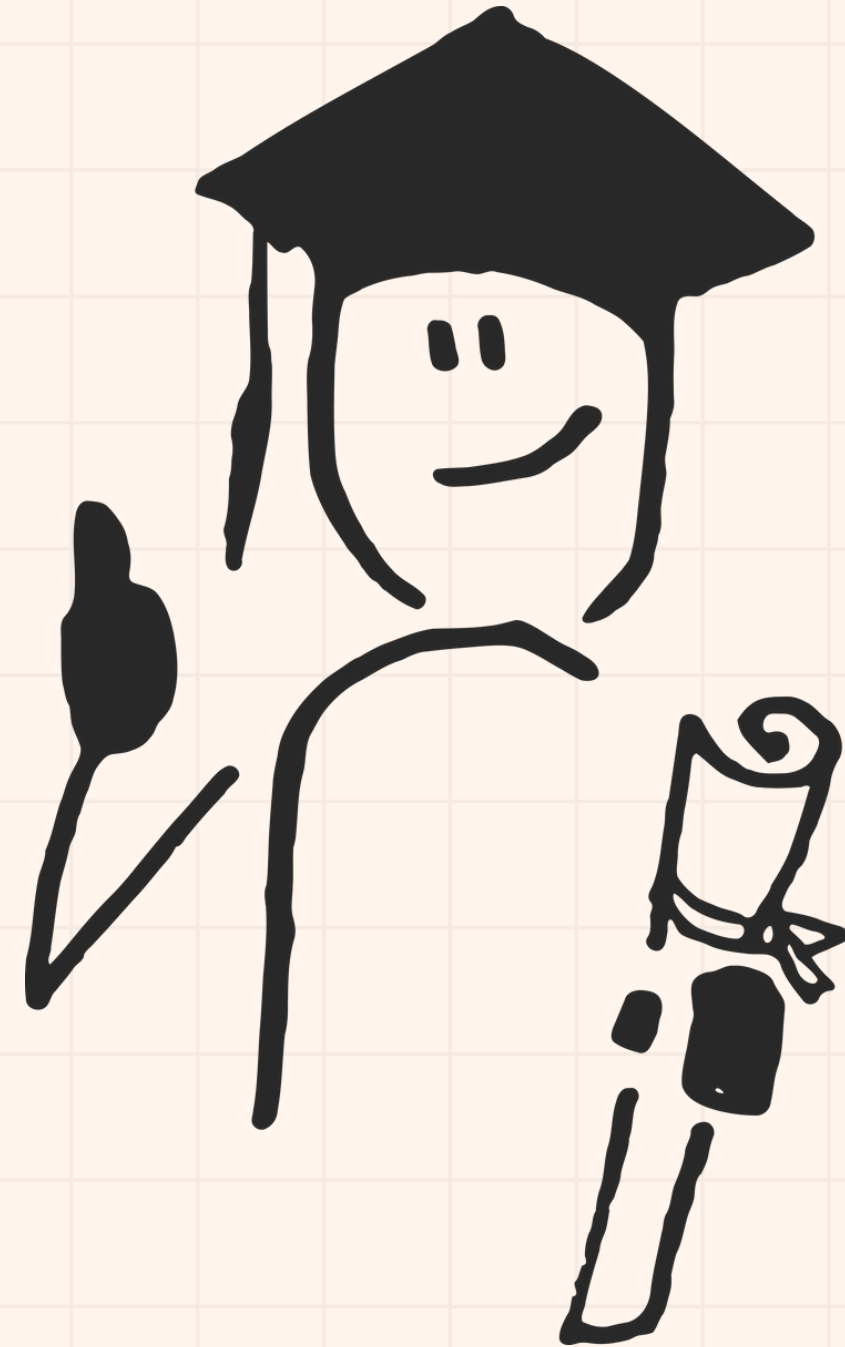
Tho Hui Yee



Tutor: THAMARAI A/P SUBRAMANIAM

INTRODUCTION

Current manual verification processes in Malaysian graduation ceremonies lead to significant delays, congestion and lineup error. To encounter these weaknesses, we propose an **Integrated Facial Recognition and QR Code System** that auto-identify verification and real-time student sequencing, ensuring a professional event experience.



PROBLEM STATEMENTS

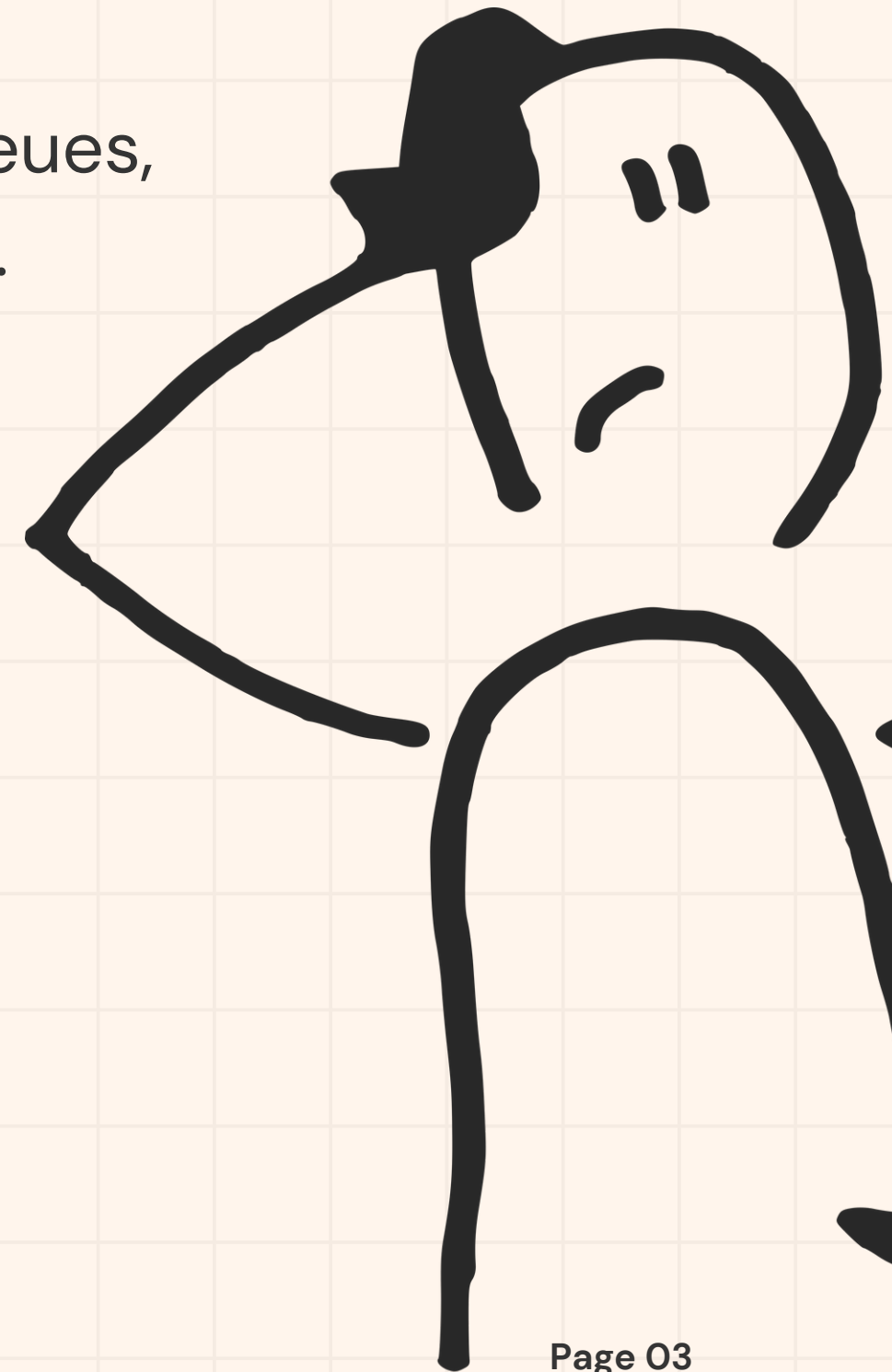
Traditional manual checks and printed lists cause long queues, staff burnout, and critical lineup error (Nordin et al., 2020).

CORE ISSUES

These methods cause long queues, crowd congestion, and event delays due to staff having to manually verify thousands of students and repeatedly troubleshoot failing scans.

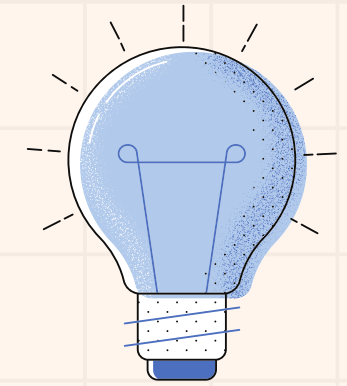
CRITICAL IMPACT

The lack of real-time automation severely hinders precise lineup organization, leading to last-minute confusion, mismatched sequencing, and a disruptive commencement flow. This compromises the professional integrity of the event.





OBJECTIVES



To design and integrate a real-time human tracking module that ensures perfect accuracy in identifying and guiding verified students to their exact pre-assigned lineup positions.

To develop and implement a functional integrated facial recognition and QR code-based student verification platform that provides highly accurate and instant identity verification.

To increase the operational efficiency of the ceremony by significantly reducing the student queuing and verification time compared to the existing manual method.

HOG + DLIB

Histogram of Oriented Gradients (HOG)

Detects face by analyzing edge directions and gradients in an image. It captures shape and facial structure

ADVANTAGES	DISADVANTAGES
• Robust to lighting changes • Works efficiently on CPU	• Sensitive to face pose • Slow for high-resolution images

HOG + DLIB 2

Dlib

uses a deep **ResNet-based model** (modified ResNet-34) that generates a **128-dimensional** face embedding for identity recognition.

ADVANTAGES	DISADVANTAGES
• High face recognition accuracy • CPU optimized and lightweight • Easy to implement with Python	• Sensitive to face pose • Slow for high-resolution images • Not ideal for highly dynamic scenes.

RETINAFACE + FACENET 512

RetinaFace: A fast face detector that also finds key facial points for good alignment.

FaceNet (512D): Turns the face into a 512-number feature code, which is compared using cosine similarity to check identity.

ADVANTAGES

- **Top Accuracy:** Superior recognition from 512D embeddings.
- **Robust Detection:** RetinaFace handles pose/occlusion well.
- **Fast Detector:** RetinaFace is faster and more efficient than MTCNN.

DISADVANTAGES

- **High Cost:** Needs significant computation for 512D vectors.
- **Large Models:** Requires more memory; harder for edge devices.
- **System Latency:** Two separate models add overall processing time.

MTCNN + FACENET

MTCNN (detectors) detects, aligns faces and predict five precise facial landmarks.
FaceNet (embedders) extracts distinctive face features for accurate identity matching.

ADVANTAGES	DISADVANTAGES
• High Accuracy: Produces clear and reliable face features for recognition. • Good Alignment: MTCNN aligns faces well, even with slight head tilts. • Easy Comparison: Face features are simple to match for verification or search.	• Slower Process: Uses two models (face detection and recognition), which takes more time. • Slow Detection: MTCNN runs slower than single-stage detectors. • Hard Training: FaceNet needs complex and resource-heavy training.

INSIGHTFACE



Detection Model: RetinaFace Scans the entire image looking for patterns that resemble human faces.

← →
Recognition Model: ArcFace Analyzes the geometric and textural details of the face (distance between eyes, jaw shape, skin texture) and compresses them into a mathematical code (512-dimensional vector).



Pros:

1. Real-time efficiency – optimized CPU execution
2. Robustness to variations – ArcFace's angular margin loss
3. Simplified detection with 5 landmarks

Cons:

1. Alignment sensitive – alter to different geometric alignments
2. Struggle with small face – small model
3. Dependency – misses face or align result in failure
4. Black box – embedding vector or initial alignment which is fail

MEASUREMENT METRICS

HOG+Dlib achieve the overall highest raw accuracy (89.17%) and preprocessed increases the accuracy to 90.83%

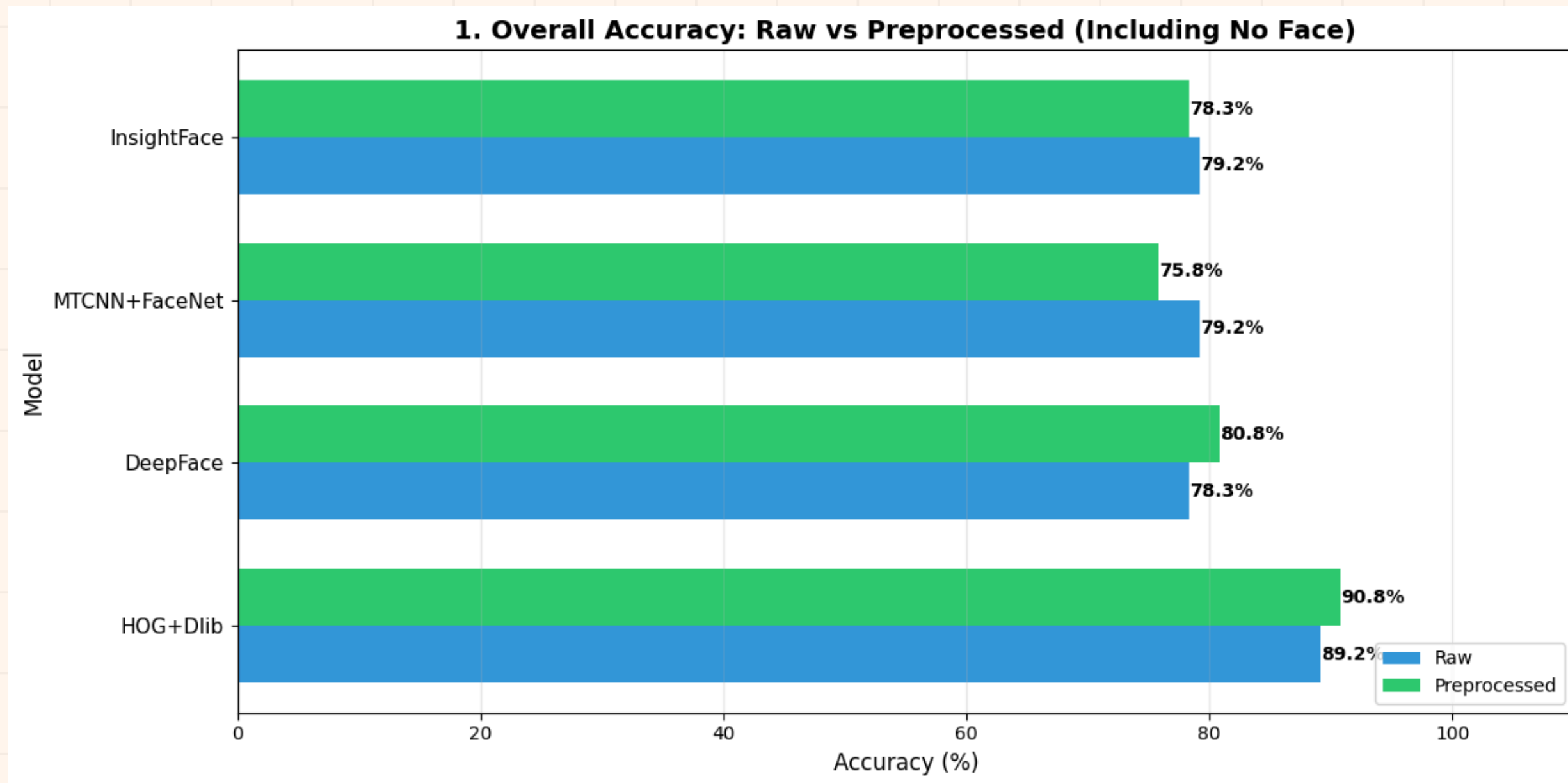


Figure 4.1 Overall Accuracy: Raw vs Preprocessed

MEASUREMENT METRICS

InsightFace achieved a perfect 0.0% FPR in both conditions, no false matches. This makes it the most reliable model for secure verification.

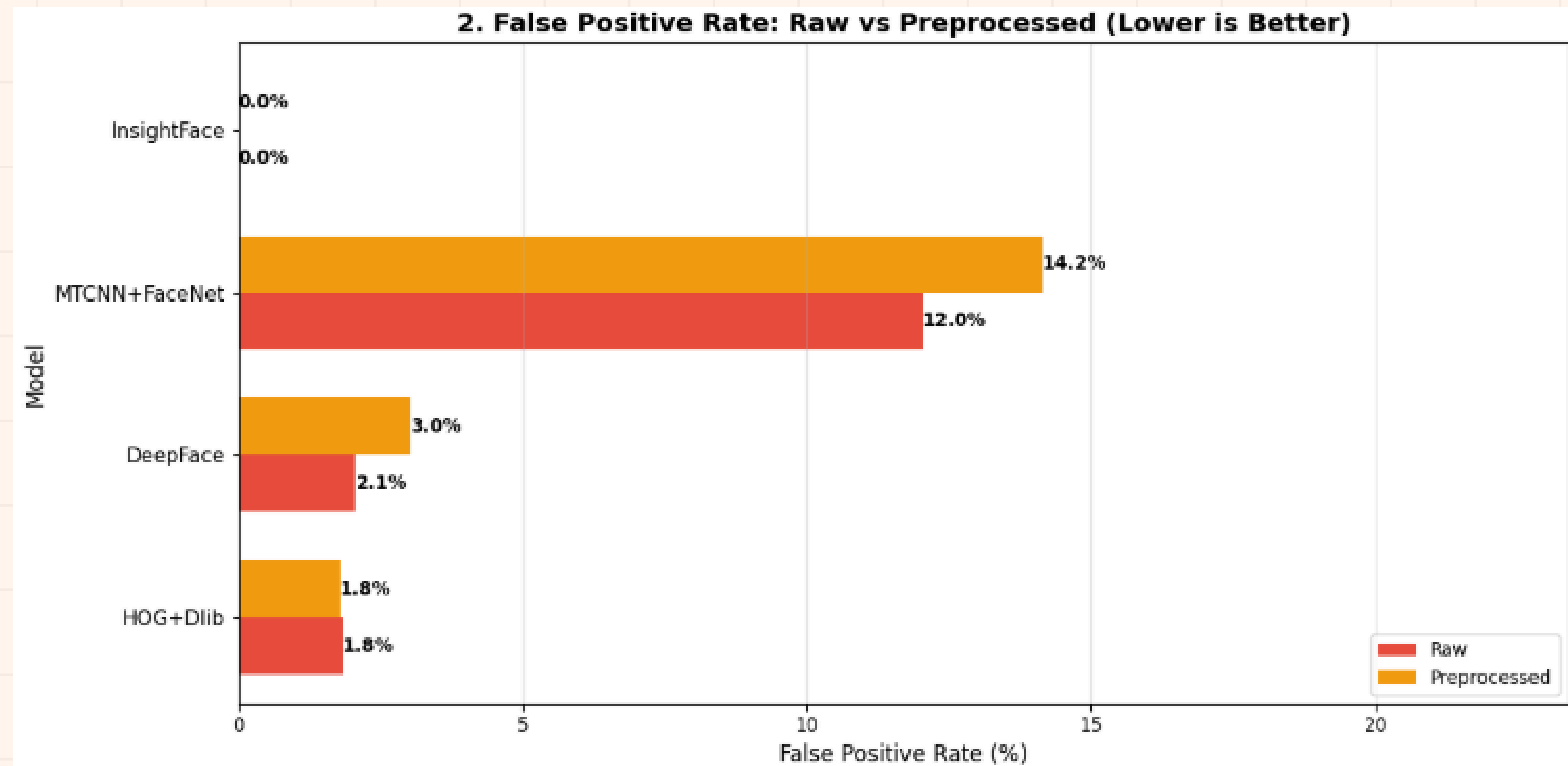


Figure 4.2 False Positive Rate (FPR): Raw vs. Preprocessed

Figure 4.3 Heatmap of original images acting on different scenarios

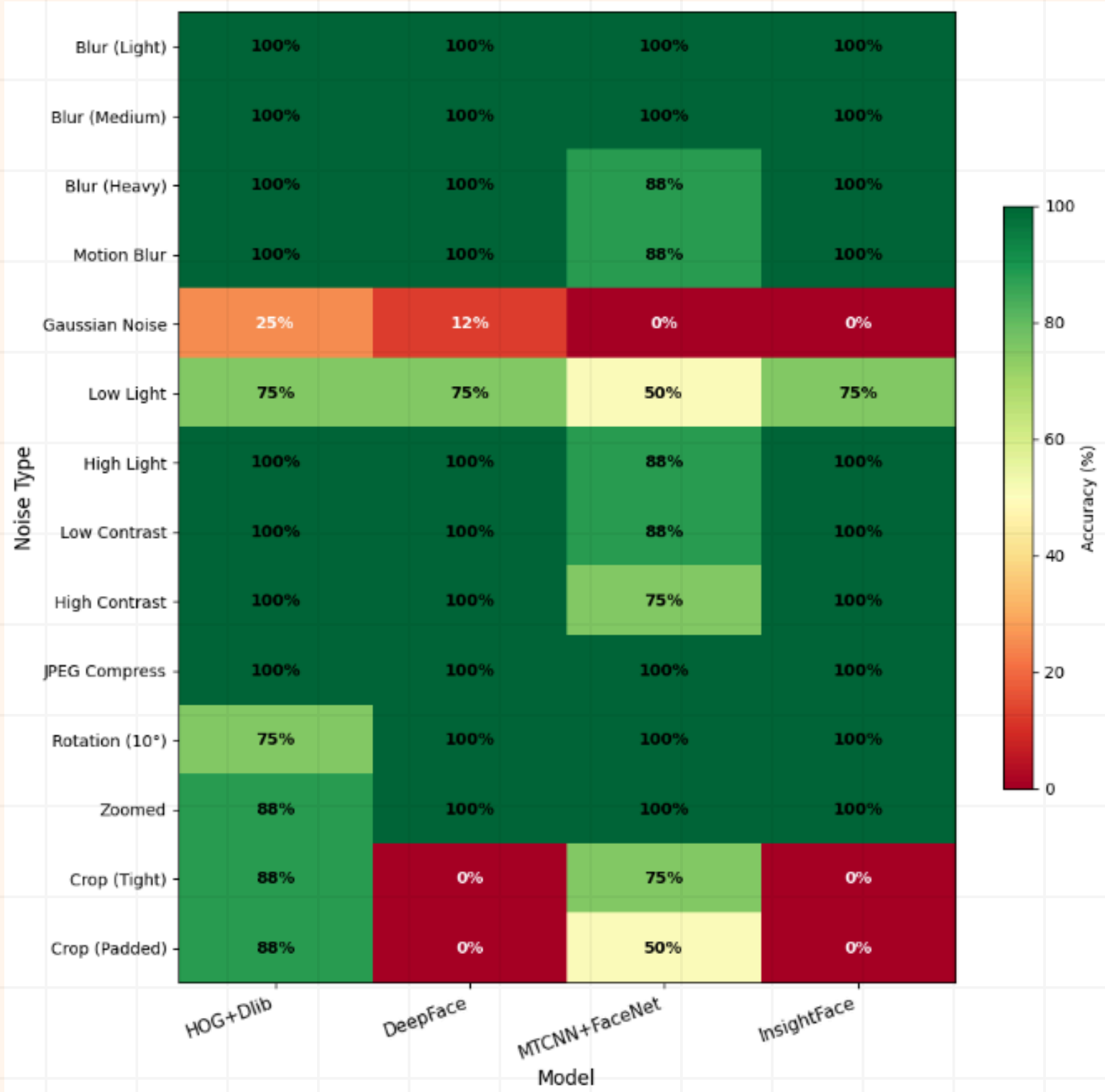
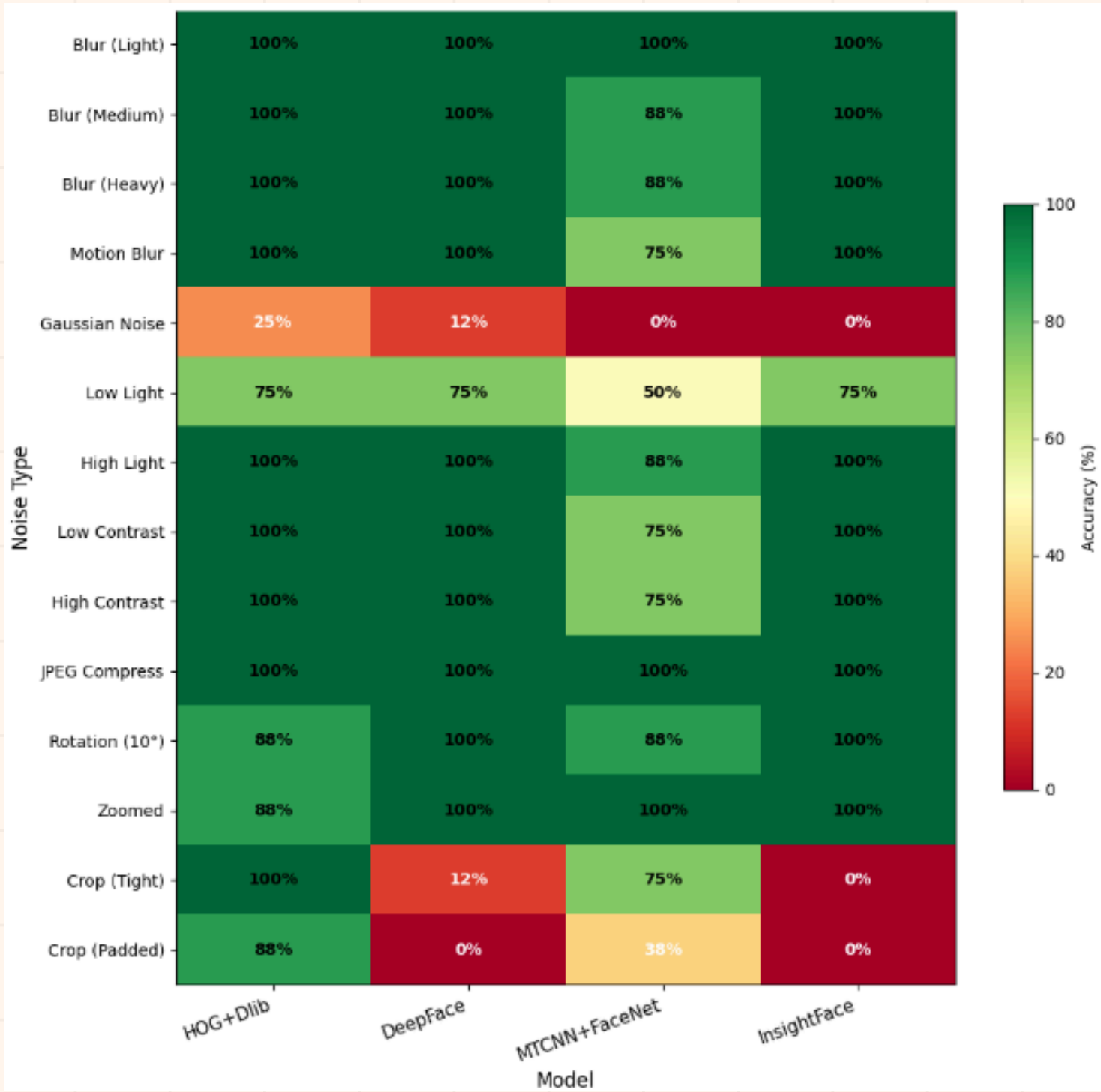
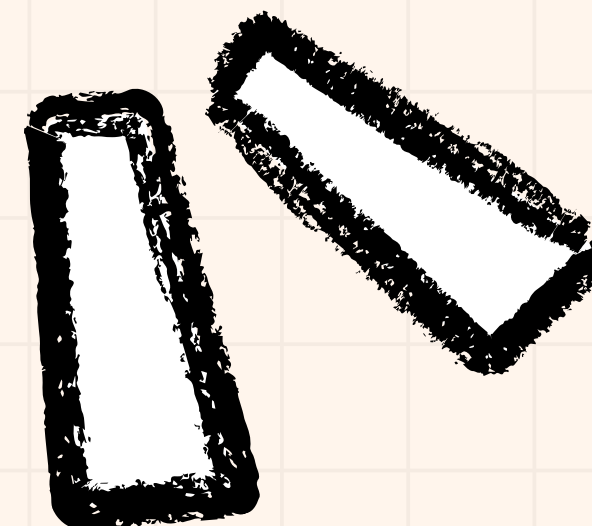


Figure 4.4 Heatmap of preprocessed images acting on different scenarios

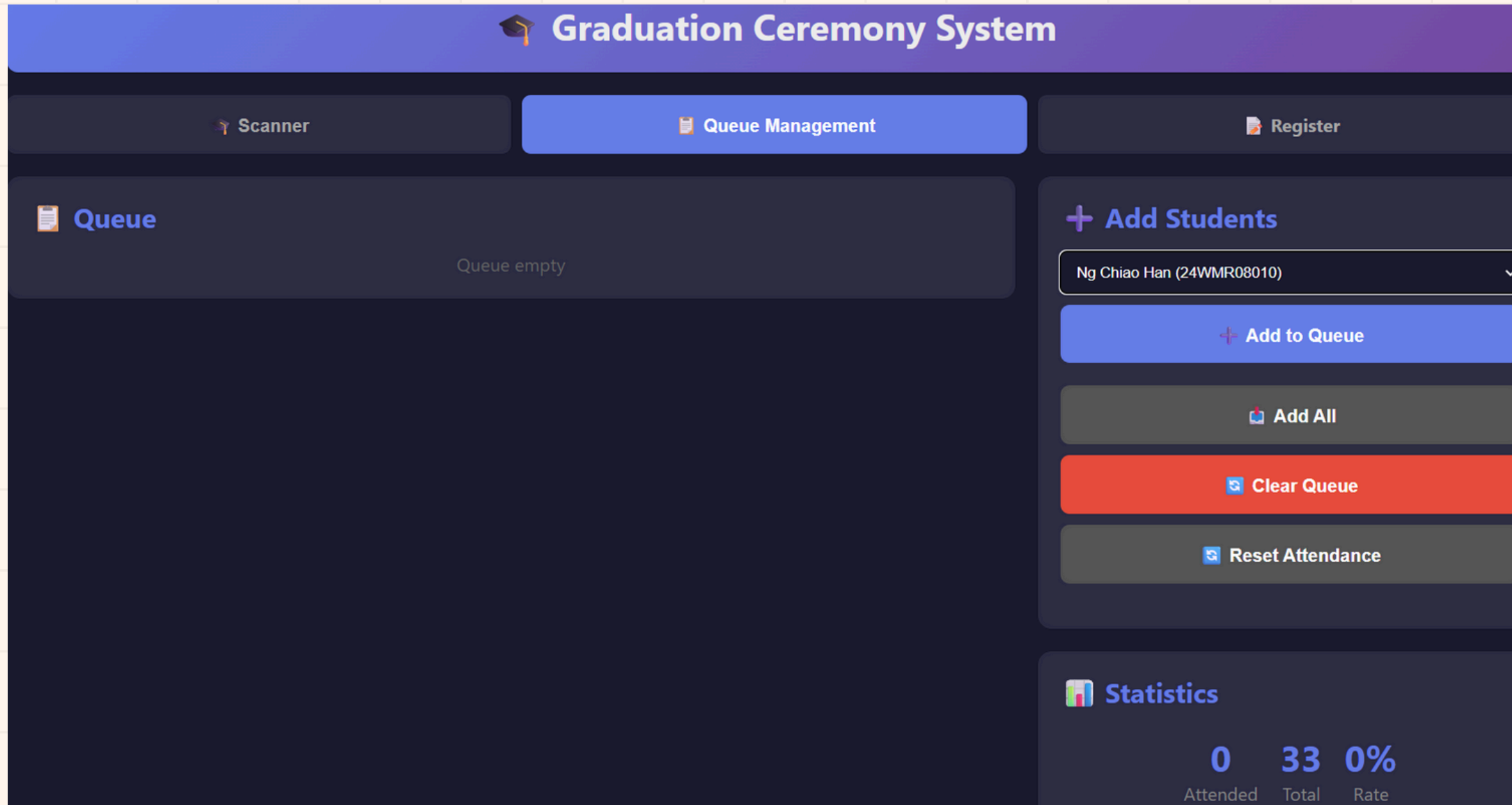




DEMO



ADD STUDENT TO QUEUE

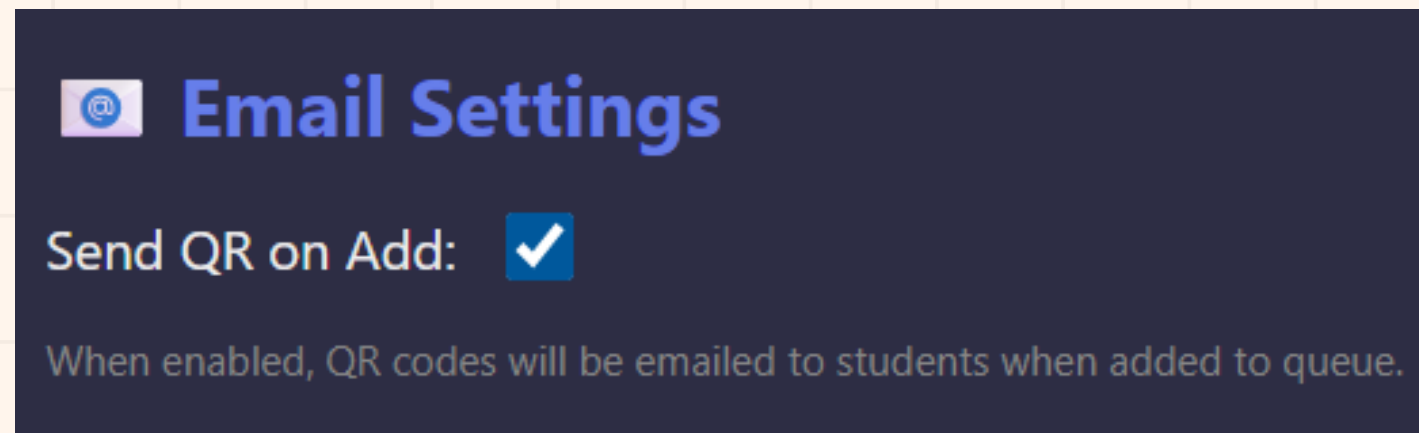


The screenshot shows the 'Graduation Ceremony System' interface. The top navigation bar is purple with a graduation cap icon and the text 'Graduation Ceremony System'. Below this is a dark blue sidebar with three main sections: 'Scanner', 'Queue Management' (highlighted in blue), and 'Register'. The 'Queue Management' section contains a 'Queue' sub-section with a document icon and the text 'Queue empty'. To the right of the sidebar is a main content area. It features a '+ Add Students' section with a dropdown menu showing 'Ng Chiao Han (24WMR08010)'. Below the dropdown are four buttons: '+ Add to Queue' (blue), 'Add All' (grey), 'Clear Queue' (red), and 'Reset Attendance' (grey). At the bottom of the main content area is a 'Statistics' section with a bar chart icon and three large numbers: '0' (Attended), '33' (Total), and '0%' (Rate).

Select a student from the drop down list then click add queue to add the student in the queue.

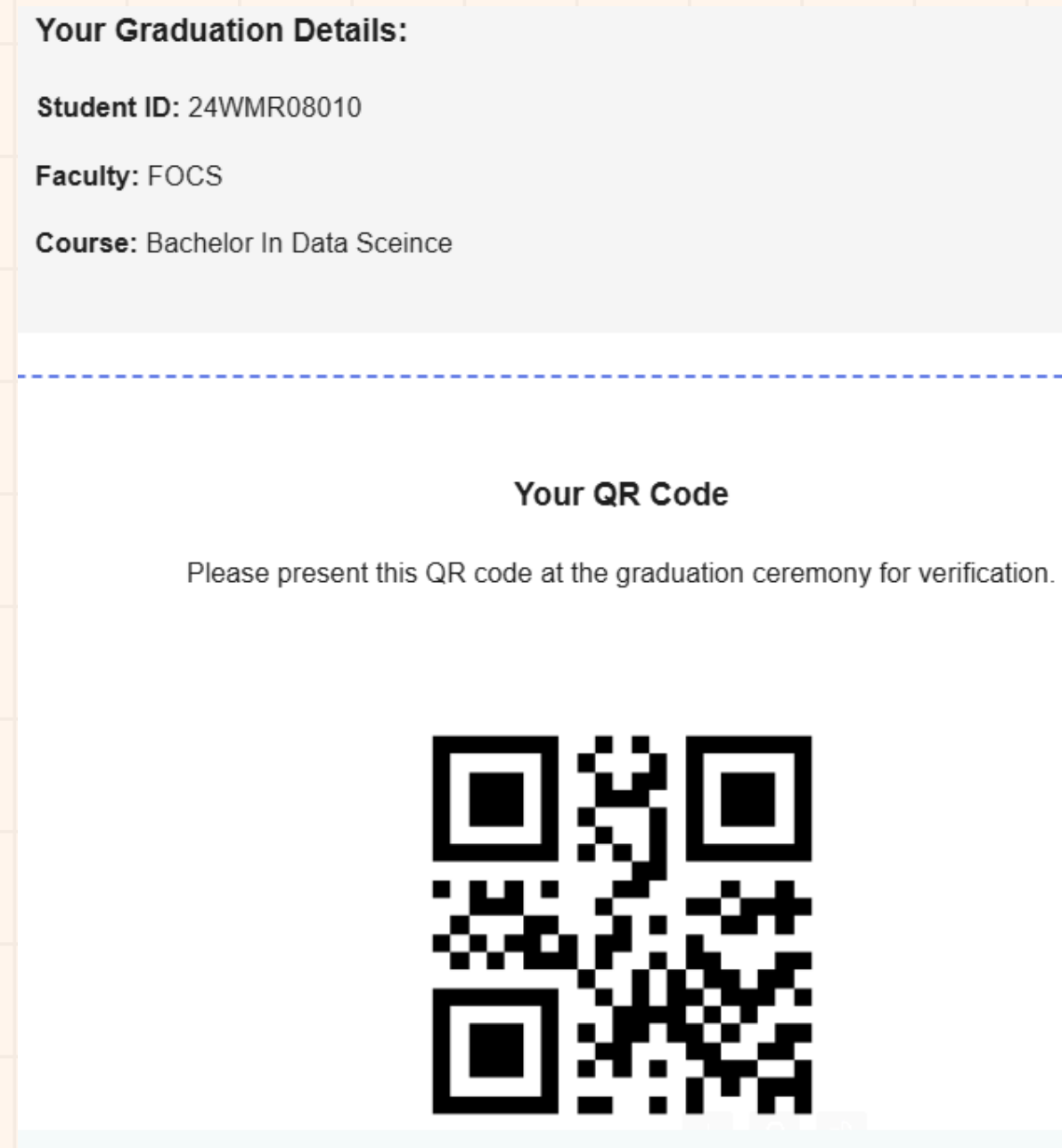
Click the add all button to add all students in queue.

QR CODE SEND THROUGH EMAIL



**When adding the students to the list,
user can allow this 'Send QR on Add'.**

**The students added to the queue will
receive an unique QR Code through the
registered email.**





SCANNER PAGE

Graduation Ceremony System

Scanner

Queue Management

Register

HOG + Dlib

Click START QUEUE to begin

Face:

QR:

Ready

Verified Student

Waiting for verification...

Controls

START

Settings

Algorithm:

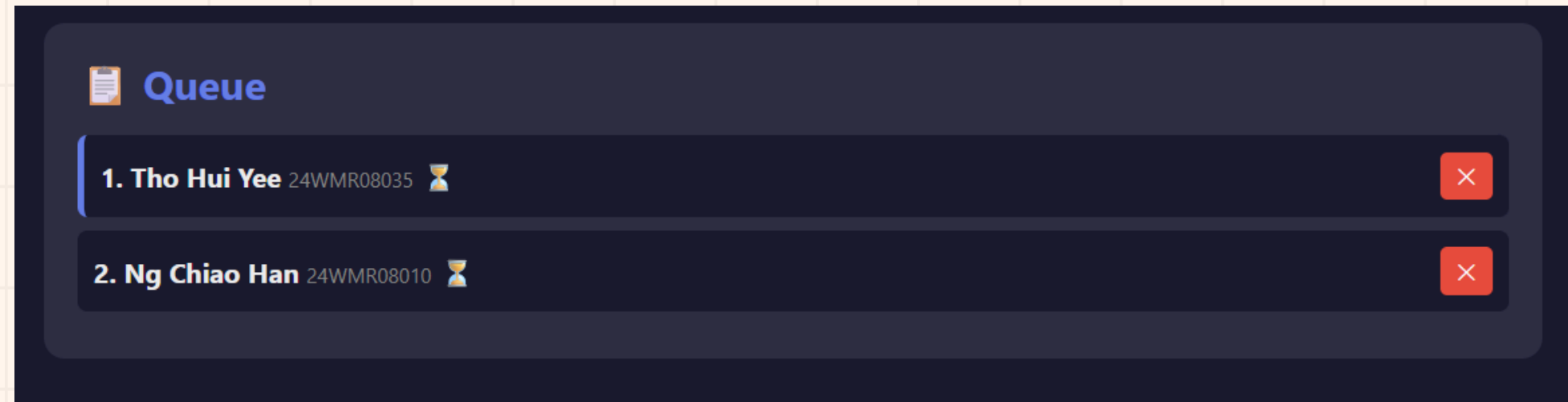
HOG + Dlib

Voice: ☒

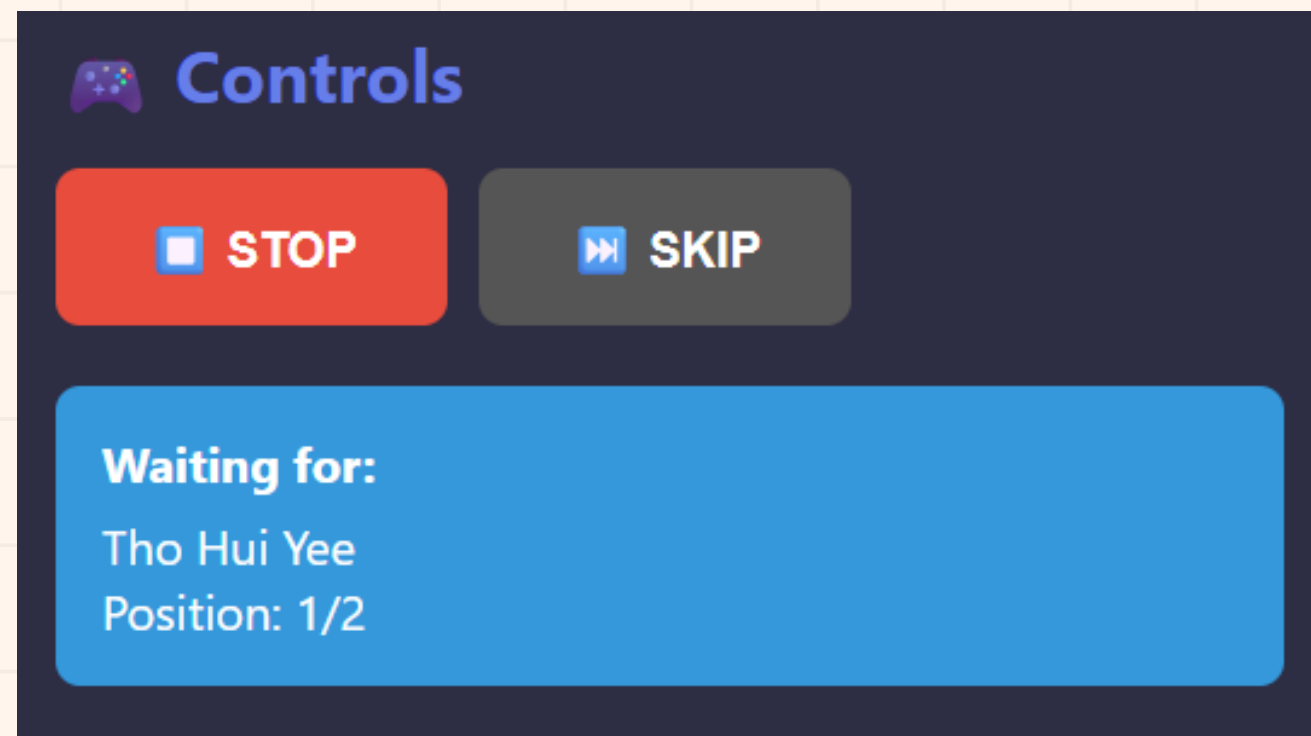
Before clicking start button,
select the algorithm from the
drop down list then click start

Student has to scan his face and
the QR code given through email
to authenticate him

VERIFICATION



The students are added to the queue in the Queue Management Page.



When the user click on the start in the Scanner Page, it will show the turn of the students.

The user can click on the 'STOP' button to stop the verification and the 'SKIP' button for skipping the current students in the queue.

VERIFICATION

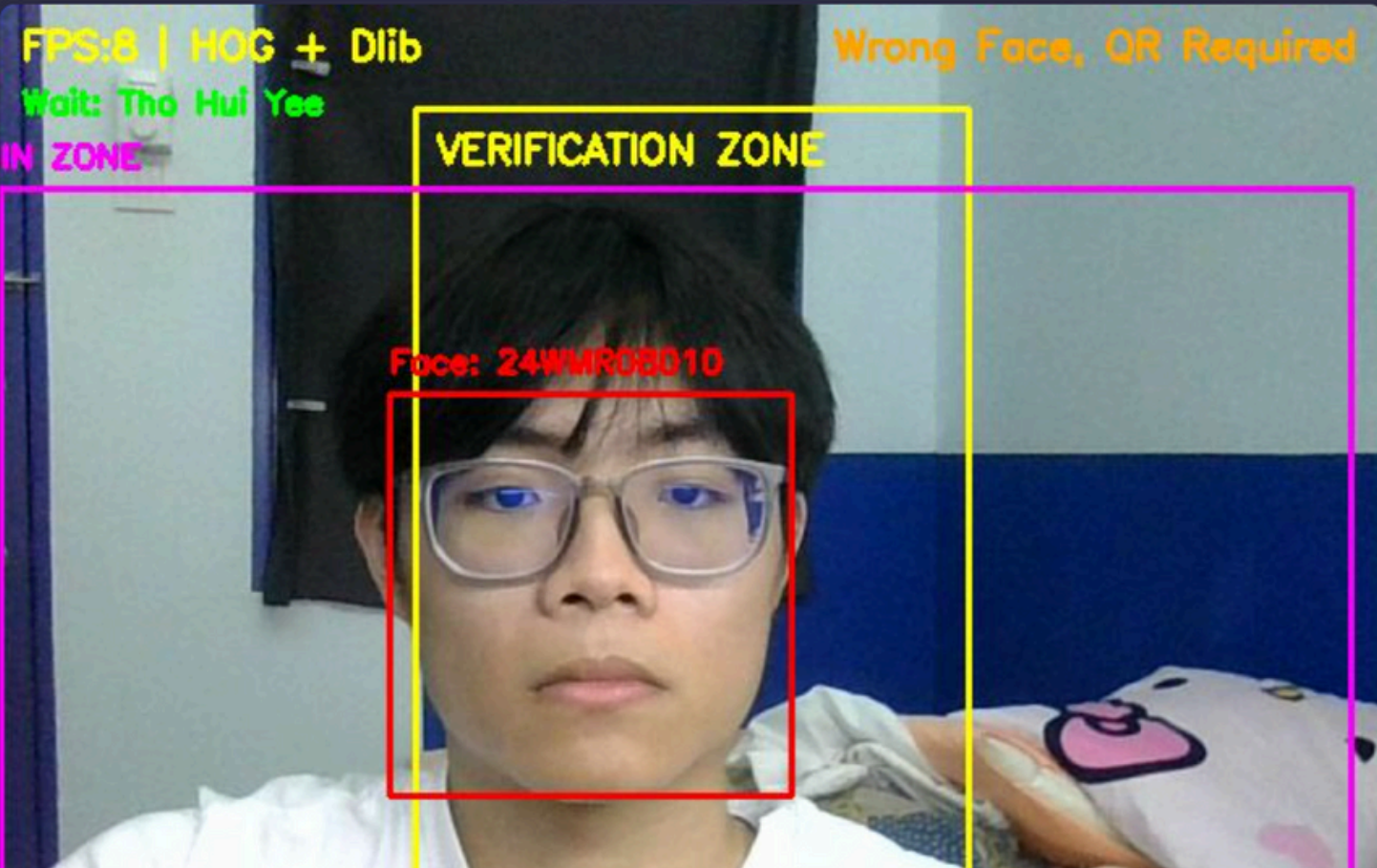
HOG + Dlib

FPS:8 | HOG + Dlib
Wait: Tho Hui Yee
IN ZONE

Wrong Face, QR Required

VERIFICATION ZONE

Face: 24WMR08010



Verified Student

Waiting for verification...

Controls

STOP

SKIP

Waiting for:
Tho Hui Yee
Position: 1/2

Settings

Algorithm: HOG + Dlib

Voice: ☒

⚠ Face: 24WMR08010 (Expected: 24WMR08035)

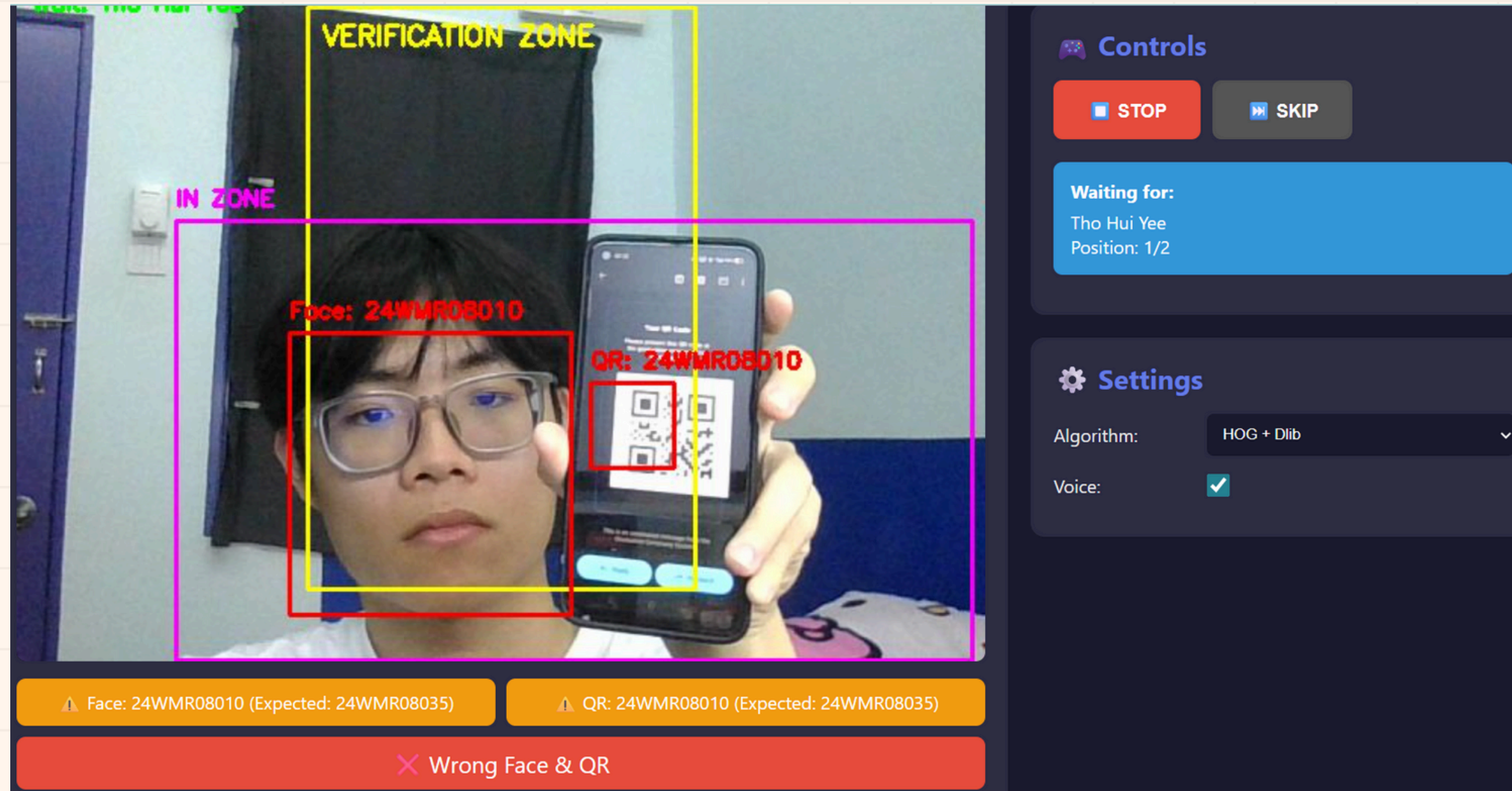
✗ No QR

Face detected, waiting QR...

The queue of the student is wrong.
It detected the face of student
24WMR08010, but expected
student is 24WMR08035.

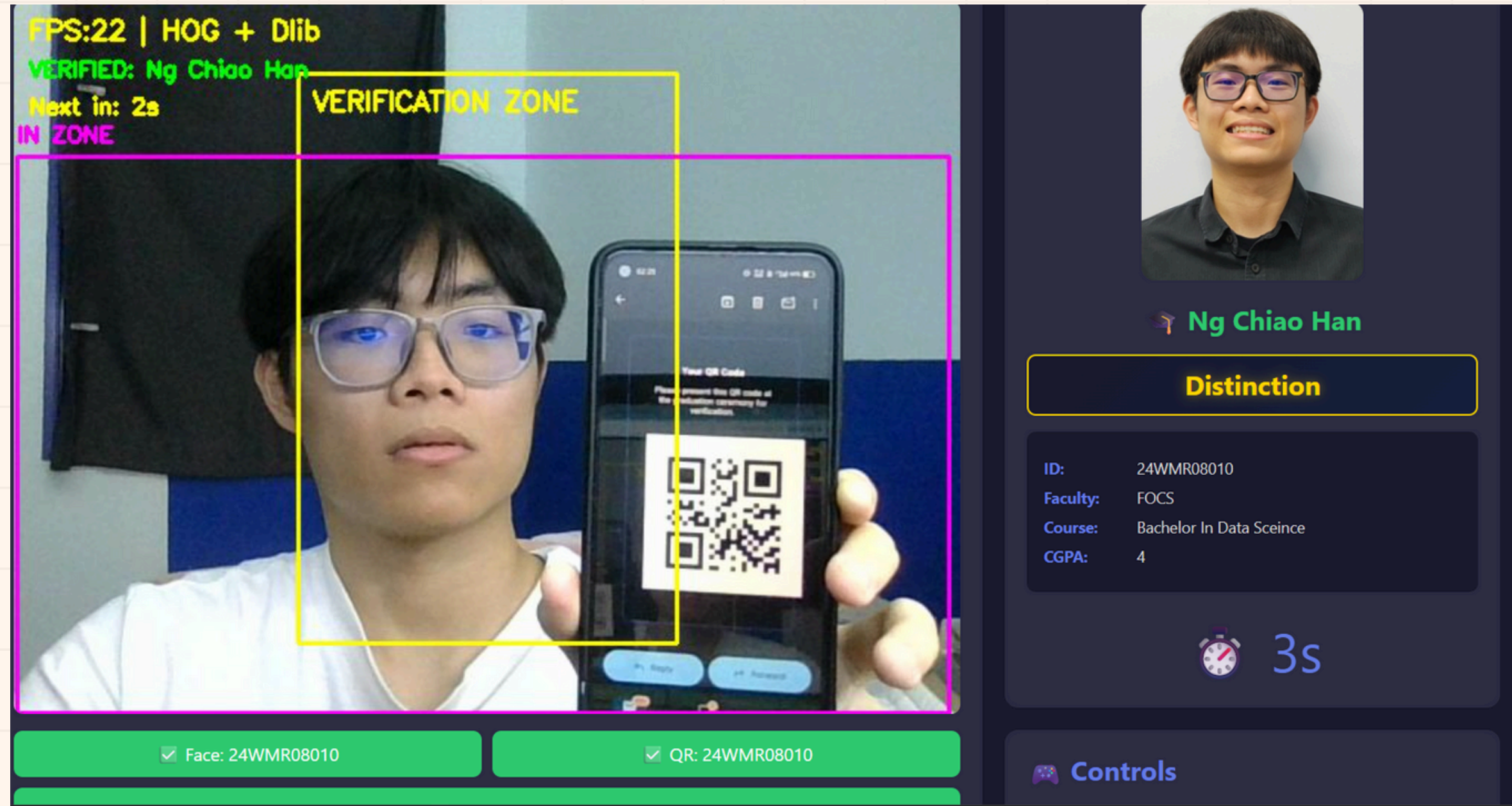
The students should show their
face and QR Code together in
order to verified correctly.

VERIFICATION



The students who stands on the wrong position and also with the wrong QR code.

VERIFICATION



The interface displays a live video feed of a student, Ng Chiao Han, holding a smartphone with a QR code. The student's face is framed by a yellow box labeled "VERIFICATION ZONE". The phone screen shows a QR code and the text "Your QR Code" and "Please present this QR code at the verification terminal for verification". The interface also shows a timer "Next in: 2s" and a status bar at the bottom with "Face: 24WMR08010" and "QR: 24WMR08010".

FPS:22 | HOG + Dlib
VERIFIED: Ng Chiao Han
Next in: 2s
IN ZONE

VERIFICATION ZONE

Ng Chiao Han

Distinction

ID: 24WMR08010
Faculty: FOCS
Course: Bachelor In Data Sceince
CGPA: 4

3s

Controls

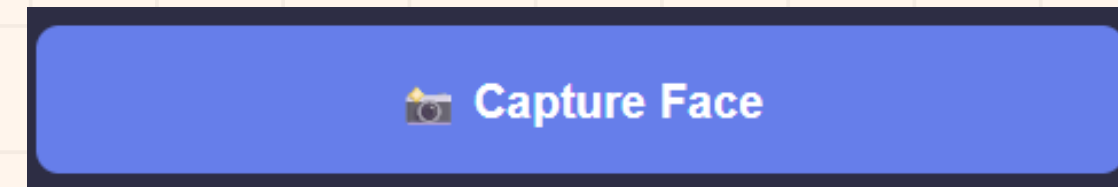
✓ Face: 24WMR08010
✓ QR: 24WMR08010

When the students show the correct face and QR Code (correct queue), it will display the details of the students with the sound.

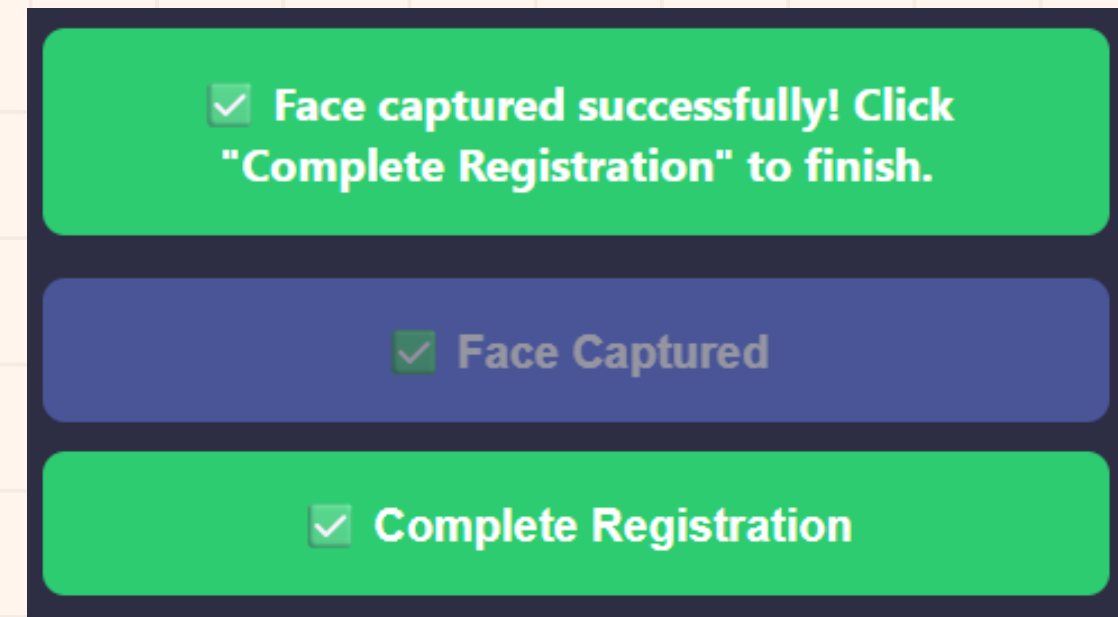
USER REGISTRATION

The user should input all the required fields before capturing their face.

After the user input, the 'Capture Face' button will change colour to allow the user to capture their face.



After captured successfully, the user should click on the 'Complete Registration' button for registration.



← Back to Scanner

Student Registration

1 Fill Details

2 Capture Face

3 Complete

Student Information

Student ID *
e.g., 12345678

Full Name *
e.g., John Doe

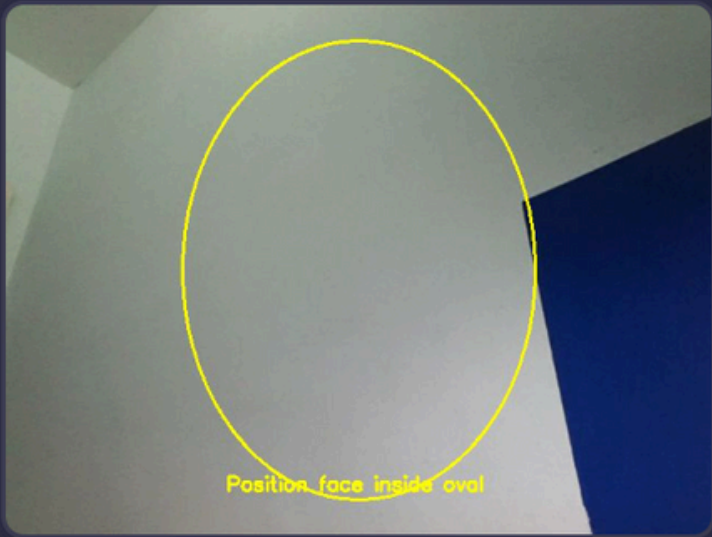
Email *
e.g., john@student.edu

Faculty *
Select Faculty...

Course *
e.g., Computer Science

CGPA
e.g., 3.50

Face Capture



⌚ Fill in details first, then capture your face

📷 Capture Face

✅ Complete Registration



DELETE USER



Delete Student

Lee Zi Jia (24WMR09013)



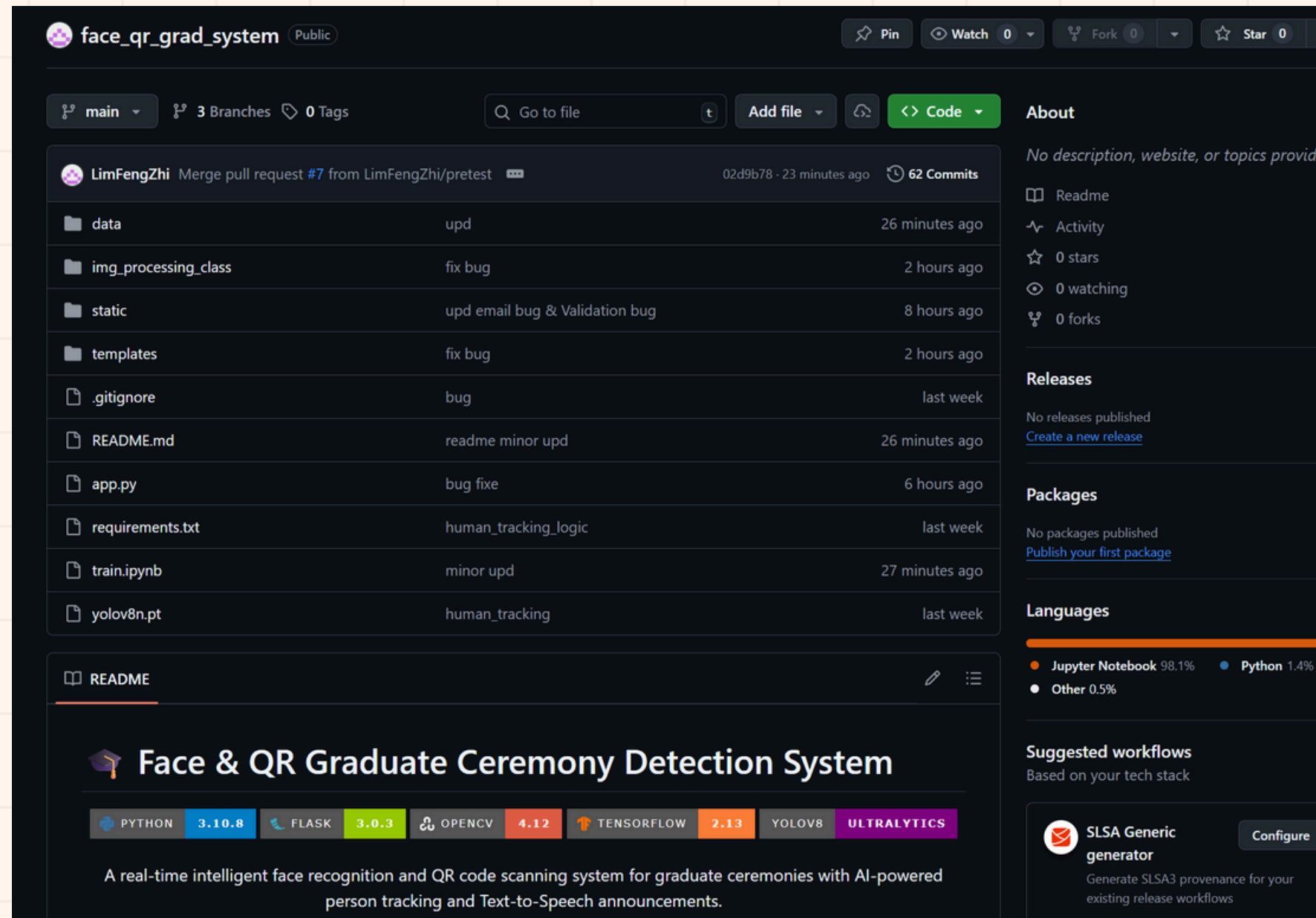
Delete Permanently



This will permanently remove the student's database record, face encodings, image, and QR code.

Select a student from the dropdown list then click the delete permanently to delete the student and its all related info in the database.

ACCESS TO OUR GITHUB





THANK YOU!

